Cecilia M. Martinez Rodriguez

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PROFESSIONAL SUMMARY

Mechanical Engineering student at Columbia University with hands-on experience in building decarbonization, off-grid solar systems, and high-energy astrophysics research. Skilled in energy simulation, HVAC modeling, SolidWorks, DFM, ANSYS, and Python, with a strong foundation in heat transfer, fluid mechanics, and materials. Proven communicator with a record of conference presentations and published technical reporting in sustainability and research contexts.

EDUCATION

Columbia University — New York, NY
MS in Mechanical Engineering — Expected May 2027
Concentration in Energy Systems.
Dean's M.S. Academic Excellence Fellowship 2026-2027.

Barnard College — New York, NY
B.A. in Physics, 3.88/4.0 — May 2026
Dean's List (All semesters).
Relevant Courses: Mechanics of Solids, Fluid Mechanics, Heat Transfer, Materials and Manufacturing, Decarbonizing Buildings Studio, and Alternative Energy Resources and Building Energy Modeling and Simulation, etc.

WORK EXPERIENCE

Jordan Energy & Food Enterprises LLC — New York, NY
Clean Energy Analyst Intern - Solar Project Development — Sept 2026 - Present
- Growing development team on active New York projects across solar system modeling, project financial analysis, and portfolio data management.

Let's Share the Sun Foundation — Adjuntas, PR
Summer Intern — Jul 2026 - Aug 2026
- Authored LSTS' inaugural 15-page Impact Report, communicating metrics and highlights to potential donors.
- Designed a branded Sheila's Hearts of Hope wooden heart keepsake for national campaign.
- Partnered with executive leadership to host and manage an immersive corporate volunteer visit, overseeing site logistics for off-grid solar and battery installations at community homes.

Building Energy Research Lab - Columbia University SEAS — New York, NY
Undergraduate Researcher — May 2025 - Aug 2025
- Integrated health outcomes as a quantifiable metric in building decarbonization planning by modeling HVAC filtration impacts on indoor air quality and energy performance across an existing 8-building residential block in Harlem.

High Energy Astrophysics Research - Barnard College / VERITAS / CTA Group — New York, NY
Undergraduate Researcher — May 2023 - May 2025
- Conducted research on high-energy astrophysics under Prof. Reshmi Mukherjee in the VERITAS/CTA group, specializing on analysis of blazar, B2 1811+31.
- Operated and calibrated VERITAS Gamma Ray Observatory during week-long shift in Arizona in April 2025, including dawn and dusk grounds surveys, 30-minute pointed observation periods, maintaining controlled shift logs and records.
- Presented findings at Lida Orzeck '68 SRI (2023, 2024, 2025); Astrofest (2023, 2024); and MARLAM (2024).
- Wrote on B2 1811+31 flare event with manuscript in progress.

SKILLS & INTERESTS

- Technical: Python; Command Line Interface; Matlab, SolidWorks DFM & technical drawing; Ansys; Microsoft Office (Excel, Word, PowerPoint); Building energy modeling.
- Physics & Instrumentation: Telescope operation and calibration; high-energy astrophysics data analysis.
- Languages: Fluent in Spanish (fully bilingual and biliterate), Conversational Catalan.
- Interests: Latin and Afrocuban dance.